Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 1 of 8 · Question 8 part b.i · 4 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

8(a)(i)	broken down in liver; by deamination; to form urea; used to make new protein/amino acids;		**(3)**

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	poor/restricted movement of synovial joints pain in joint; backflow of blood in heart; poor/inefficient blood circulation/oxygenation; less energy/tired;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
8(b)(i)	• not sex-linked; • more females affected than males/more males would be affected; • only females would be carriers (if sex-linked); • male 1/3/11 would show condition if sex-linked; • but these males are carriers;		**(4)**

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	• probability that child has condition is ¼ /1:3 / 1 in 4/25%/0.25; • probability that child is male is ½/1 in 2/1:1 / 50%/0.5; • so probability that it is affected male is 12.5%/0.125/$\frac{1}{8}$;		**(3)**

Total 13 Marks

Paper Total 90 Marks

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 1 of 8 · Question 8 part b.i · 4 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 2 of 8 · Question 8 part b.ii · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
8				
(a)	(i)	B; (an alternative form of gene) A it is not an allele C it is not an allele D it is not an allele		1
	(ii)	only expressed in the homozygous state/not expressed in the heterozygous state;	Allow valid descriptions/examples of homozygous/heterozygous states	1
(b)	(i)	grandfather (condition) × grandmother; (no condition) father (cond) × mother (no cond) → student (condition) ; brother (condition) sister (no condition);		3
	(ii)	• occurs only in males/not females/more common in males/females are carriers; • recessive; • probably sex-linked/carried on X chromosome;		3 Total 8

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 2 of 8 · Question 8 part b.ii · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 3 of 8 · Question 7 part a.i · 2 marks

2022 JAN · Paper 02 · find part a.i in this mark scheme extract

number				
7 (a)	(i)	• appears in both males and females; • in similar numbers;		2
	(ii)	• condition present in heterozygous genotype/only one allele needed/affected are carriers; • only some children (from affected and unaffected parents) are affected/not affected;		2
(b)	(i)	• probability of a boy is 0.5/50%; • probability of no Huntington's is 1.0; • overall probability is 0.5 × 1.0 = 0.5/50%/1;		3
	(ii)	person 3 must be hh/homozygous recessive; does not carry a faulty allele; person 2 is Hh/heterozygous/carries a faulty allele / carries one normal allele; as father not affected; offspring either Hh or hh; half with condition and half without/50% chance of being affected;	marks can be awarded for suitable genetic crosses that are annotated	6

Total 13 marks

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Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 4 of 8 · Question 7 part a.ii · 2 marks

2022 JAN · Paper 02 · find part a.ii in this mark scheme extract

number				
7 (a)	(i)	• appears in both males and females; • in similar numbers;		2
	(ii)	• condition present in heterozygous genotype/only one allele needed/affected are carriers; • only some children (from affected and unaffected parents) are affected/not affected;		2
(b)	(i)	• probability of a boy is 0.5/50%; • probability of no Huntington's is 1.0; • overall probability is 0.5 × 1.0 = 0.5/50%/1;		3
	(ii)	person 3 must be hh/homozygous recessive; does not carry a faulty allele; person 2 is Hh/heterozygous/carries a faulty allele / carries one normal allele; as father not affected; offspring either Hh or hh; half with condition and half without/50% chance of being affected;	marks can be awarded for suitable genetic crosses that are annotated	6

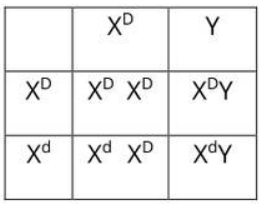
Total 13 marks

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Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 5 of 8 · Question 8 part b.ii · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

8 (a)	any 4 from • change in base sequence produces different mRNA; • altered codon (or codons); • so different amino acids coded; • incorrect protein made; • could produce a stop codon so no protein made/faulty protein made;		4
(b) (i)	• faulty allele present on one chromosome only; • effects overridden by a dominant allele;		2
(ii)	• one male in generation 1 has DMD; • (faulty) allele must be on X chromosome; • male is XY/does not carry dominant allele on Y chromosome; • carriers present;		3
(iii)	• correct parental genotypes; • correct genotypes for generation 1; 		2
			Total 11

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 5 of 8 · Question 8 part b.ii · 3 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 6 of 8 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · find part b.ii in this mark scheme extract

3 (a)	(i)	nucleus;		(1)
	(ii)	allele;		(1)
	(iii)	too many fingers/toes;		(1)
(b)	(i)	heterozygous/heterozygote/carrier;		(1)
	(ii)	Y;		(1)
(c)		Any four from: • double helix/two strands; • consists of nucleotides; • four bases; • each strand held together by hydrogen bonds/bases linked by hydrogen bonds; • complementary base pairing/A-T, G-C;		(4)

Total Question 3 = 9 marks

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 7 of 8 · Question 5 part b.ii · 4 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

5 (a)	(i)	• 13 750 ÷ 7500; • 1:1.83;	Allow 1:1.8/1:2/ 1:1.8333... allow if they do it the other way round ie 7500 ÷ 13750	2
	(ii)	• person A has an infection/disease; • Immune response/more white blood cells needed, to produce antibodies/phagocytosis/destroy pathogen/virus/bacteria;	ignore sick ignore fight infection allow alternative correct explanation	2
	(iii)	Any two from: • excess bleeding; • poor/no blood clotting; • poor/slow wound healing/more time for pathogens to enter wound;		2
(b)	(i)	• tiredness/fatigue/high heart rate/faster breathing; • less oxygen carried/transported (to body cells); • less aerobic respiration; • less energy;		4
	(ii)	• HA is dominant; • parents are heterozygous; • HA expressed in heterozygous form; • so only needs one allele;	ignore ref to allele/gene for mpt 1	4

Total Marks 14

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 7 of 8 · Question 5 part b.ii · 4 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 8 of 8 · Question 8 part b.i · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
8 (a)	(i)	B; (an alternative form of gene) A it is not an allele C it is not an allele D it is not an allele		1
	(ii)	only expressed in the homozygous state/not expressed in the heterozygous state;	Allow valid descriptions/examples of homozygous/heterozygous states	1
(b)	(i)	grandfather (condition) × grandmother; (no condition) father (cond) × mother (no cond) ; brother (condition) sister (no condition); student (condition)		3
	(ii)	• occurs only in males/not females/more common in males/females are carriers; • recessive; • probably sex-linked/carried on X chromosome;		3 Total 8

Type MS · Build or read a pedigree to decide how a condition is inherited

Answer 8 of 8 · Question 8 part b.i · 3 marks

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